Robot Learning: From Fundamentals to Foundation Models

Lecture 12: Frontier & Open Problems

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Every Week on Twitter: Robotics is Solved!

Lex Sokolin | Generative Ventures @LexSokolin · May 8
"We are approaching the endgame for robotics"

> **Genesis AI** @gs_ai_ · May 6
> We are back. After one year of quiet building.
>
> Introducing GENE-26.5, our first robotic brain that takes a major step toward human-level capability.
>
> —

Brett Adcock
@adcock_brett

we just had an AI breakthrough in our lab

robotics is about to have its ChatGPT moment

and that moment is happening tomorrow

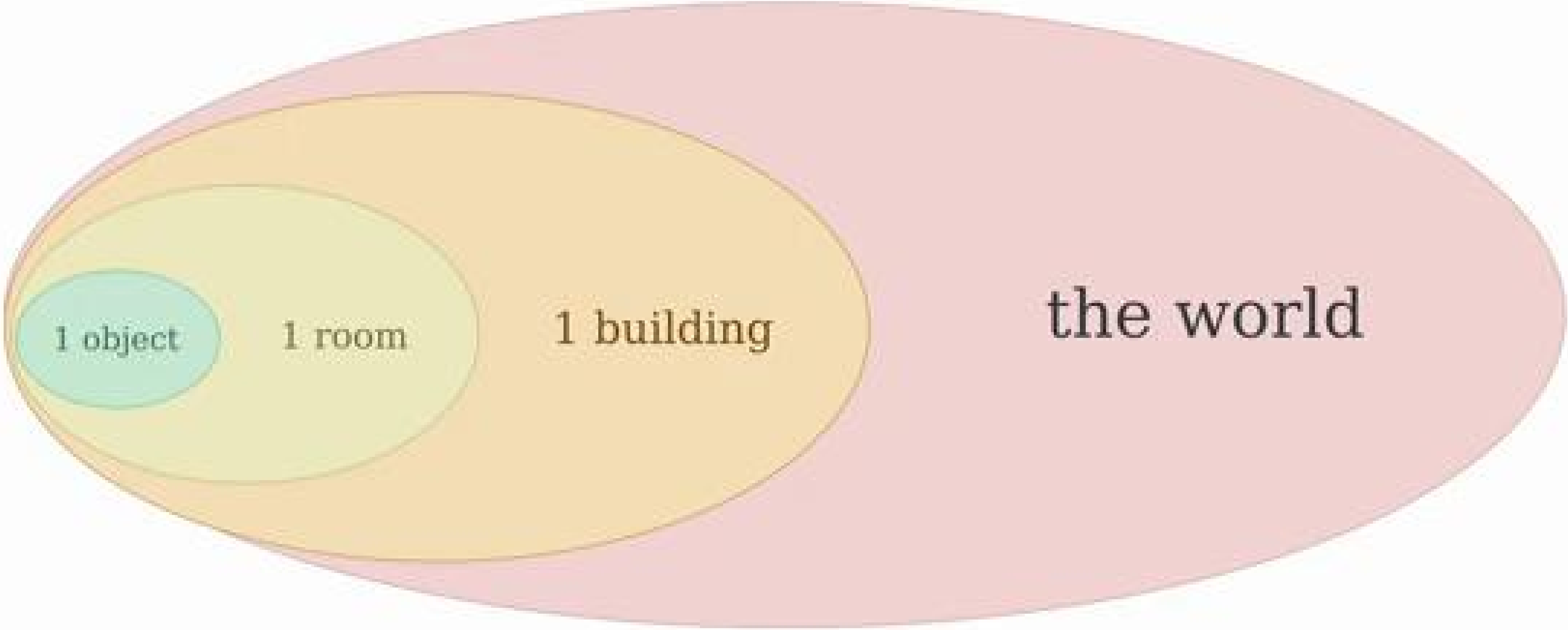
1:17 AM · Jan 7, 2024 · **2.4M** Views

> **Jim Fan** @DrJimFan · May 8
> I promise this will be the best 20 min you spend today! Robotics: Endgame, the sequel to my last year's Sequoia AI Ascent talk, "Physical Turing Test". I laid out the roadmap for solving Physical AGI as a simple parallel to the LLM success story. Be a good scientist, copy

How Autonomous Robots (Usually) Generalize

Boston Dynamics

Fu et. al., Stanford 2024


1 object
1 room
1 building
the world

The Quest for the Best of Both Worlds

Non-Embodied Foundation Models

- ✅ Internet-scale Data
- ✅ Broad Generalization
- ❌ Lacks Physical Grounding
- ❌ Non-Embodied Sensing

Specialist Embodied Models

- ✅ Physical Grounding
- ✅ Embodied Sensing
- ❌ Narrow, Task-specific Data
- ❌ Poor Generalization

The Path to Embodied Intelligence

- Today's foundation models lack physical grounding

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- Embodied data inherently **multimodal, spatial & temporal**

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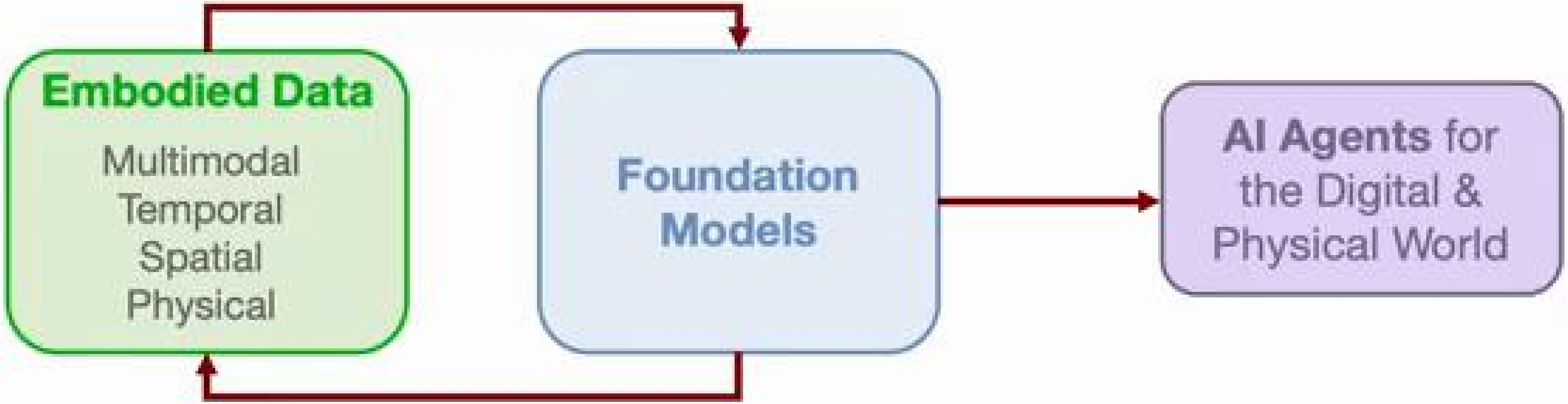


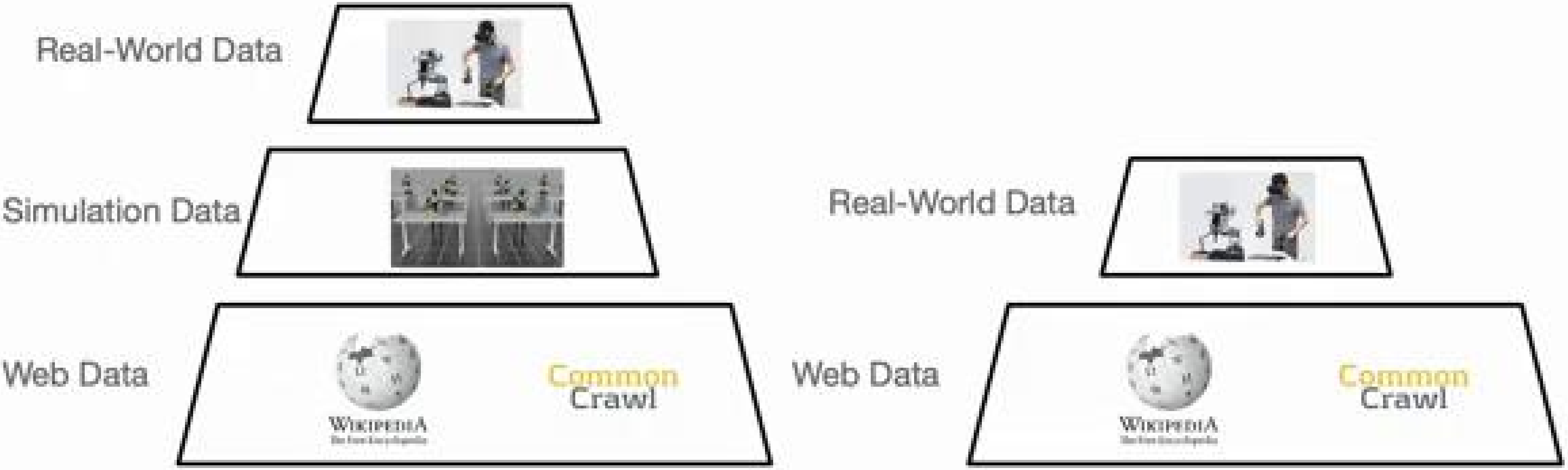
What is the Best Backbone for Robotics?

Vision-Language Model

Generative Video Model

Doesn't matter if we have enough data to train from **scratch**?

What is the Best Data Recipe?



What is the Best Data Recipe?

Data Collection Interfaces


UMI-Style

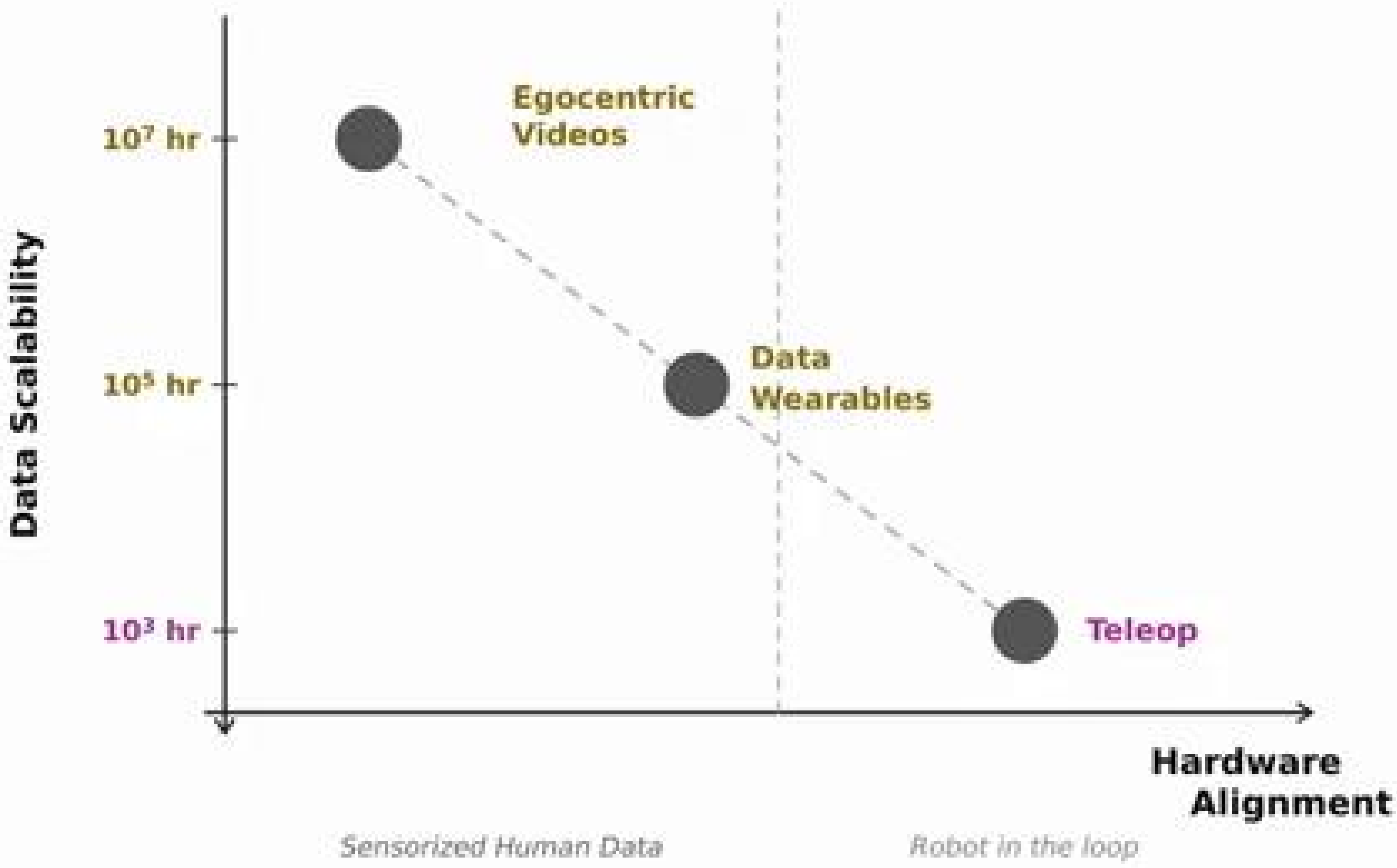

Gloves for Dexterous Hands


Static Bimanual Puppeteering


VR

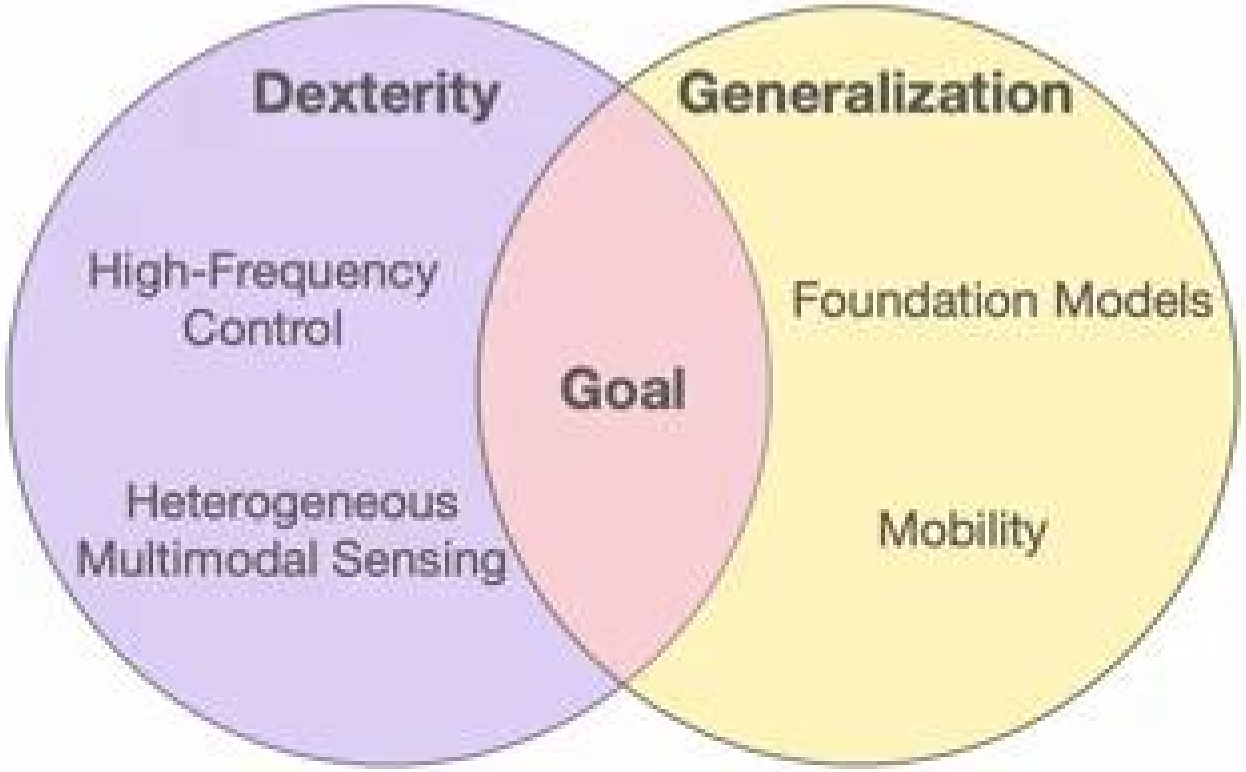

Mobile Bimanual Puppeteering

Data Scalability vs Hardware Alignment



Towards **Dexterous Generalist Policies**

Dexterity and generalization require different ingredients

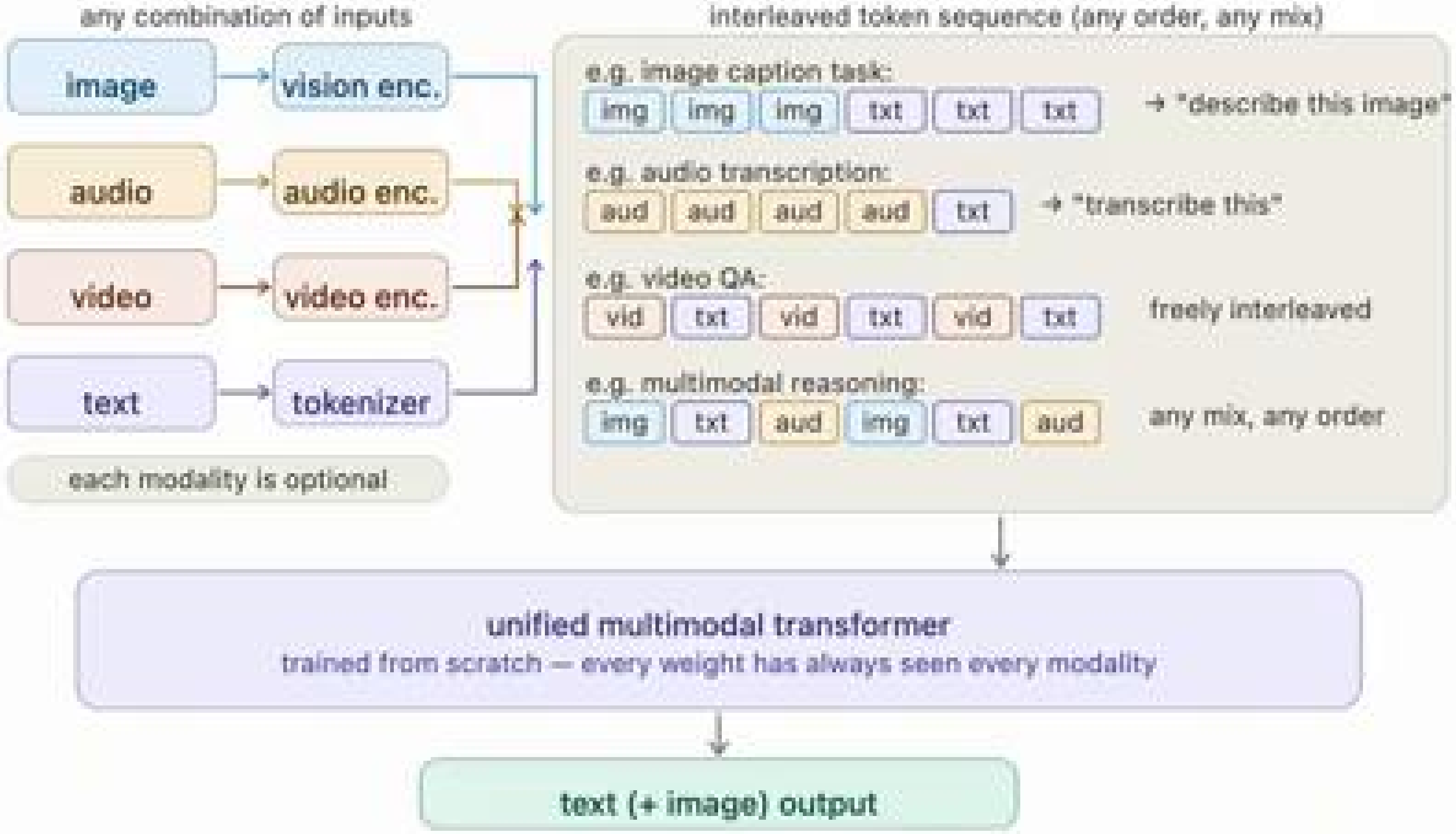


Dexterity Requires More Than Vision

Self-supervised 3D Shape and Viewpoint Estimation from Single Images for Robotics
Mees, Tatarchenko et al., IROS, 2019.

Recap: Native Multimodal Models

- Instead of adding vision to an existing LLM, train all modalities jointly from scratch

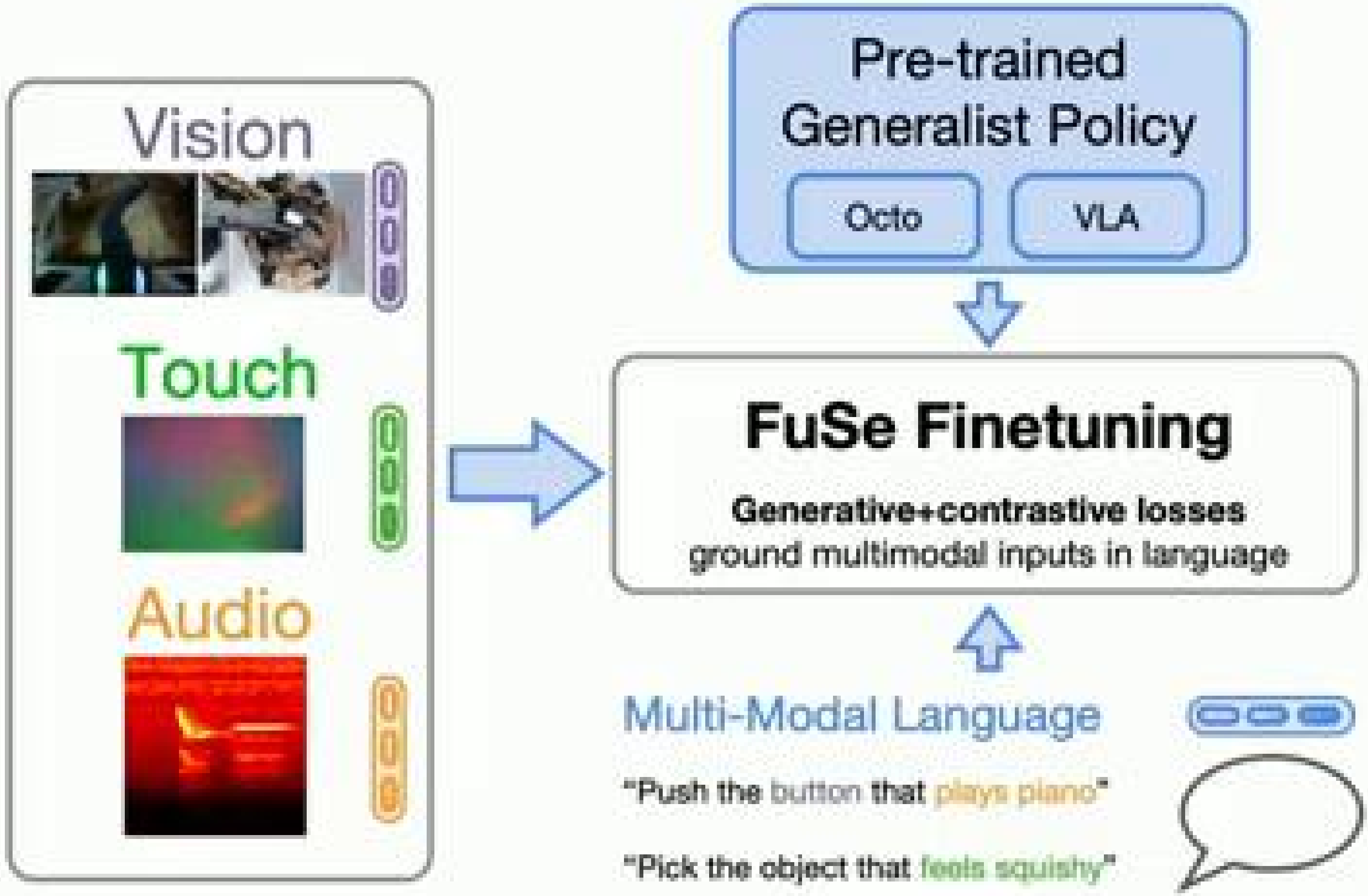


The Long Tail of Robot Sensing

- **Scarcity**: many relevant sensing modalities (e.g. touch) not available at scale
- **Pairing**: cross-modal paired data (e.g. RGB + depth+ tactile + force simultaneously) is nearly nonexistent
- **Heterogeneity**: sensors vary across robot embodiments, making transfer hard even when data exists

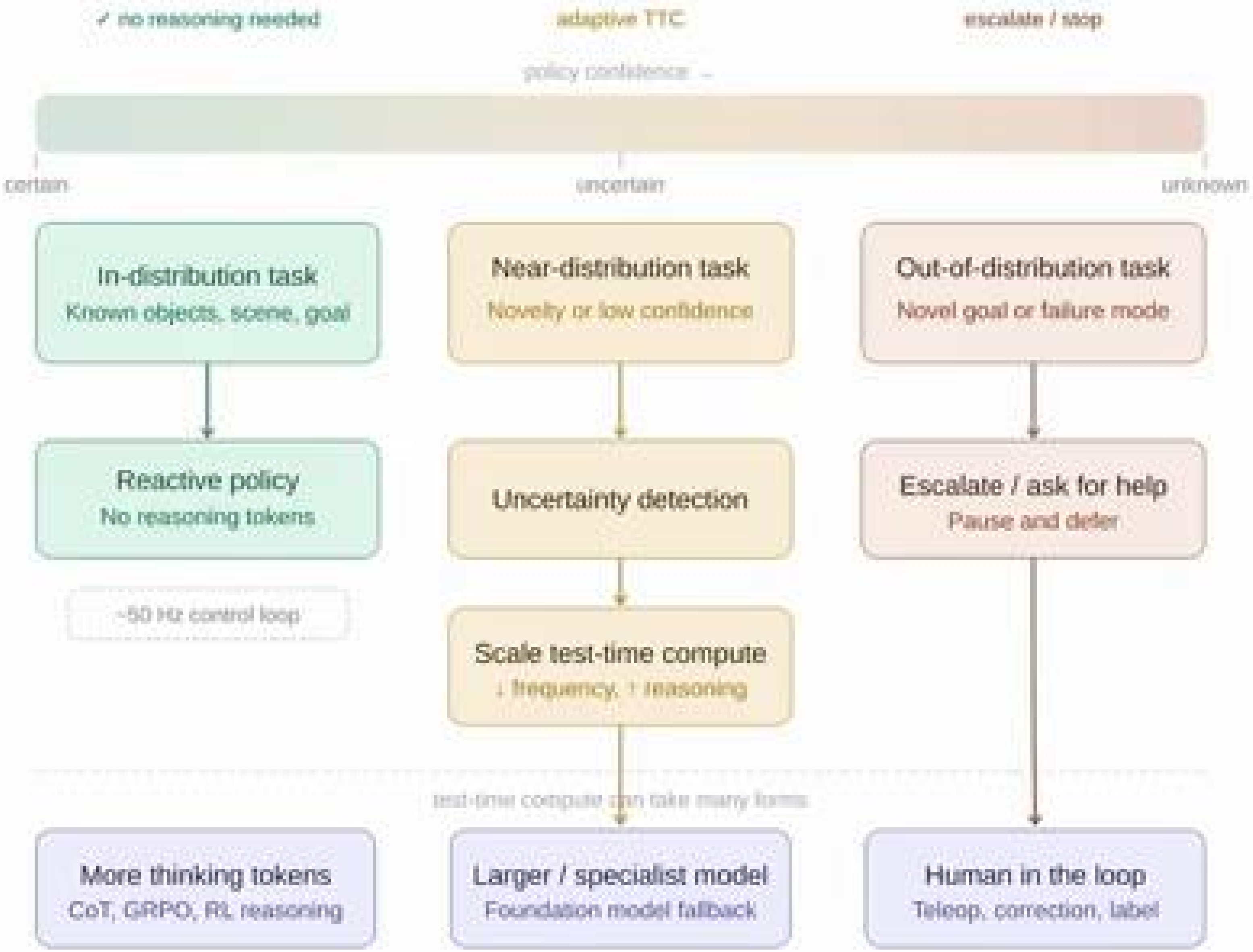
Reason Across **New** Modalities?

Giving VLAs Senses They Were Never Trained On



Beyond Sight: Finetuning Generalist Robot Policies with Heterogeneous Sensors via Language Grounding
Jones*, **Mees***, Sferraza*, Stachowicz, Abbeel, Levine. ICRA, 2025.

When & How to Reason Intelligently?



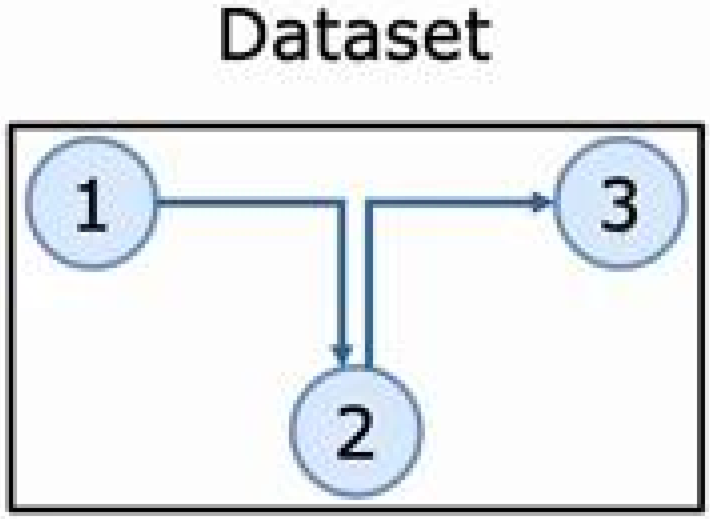
How does a model know what it doesn't know?

Generalization across many axes: objects, environments, embodiments, instructions, tasks…

Open Problem: Introspection

Current Robot Models Rely on Imitation Learning

Policy bounded by the demonstrations in the dataset



Can we Scale RL for Robotics?

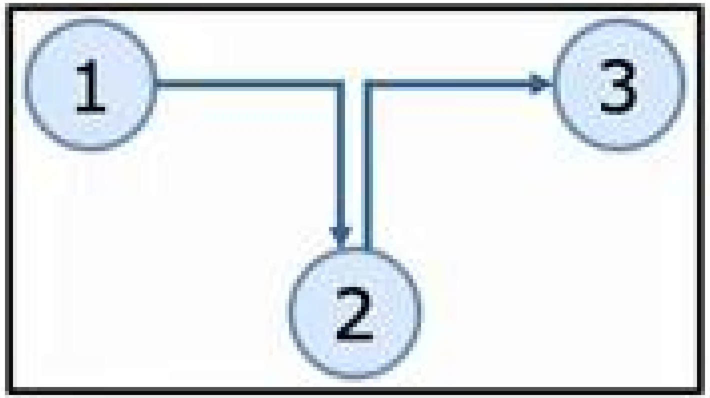
Imitation learning:

- Imitates seen behaviors in the data

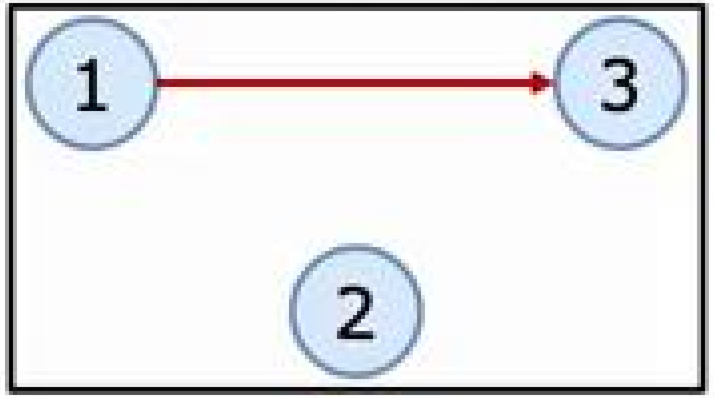
Reinforcement learning (RL):

- Learns near optimal policies from suboptimal data
- Stitches suboptimal trajectories

Dataset

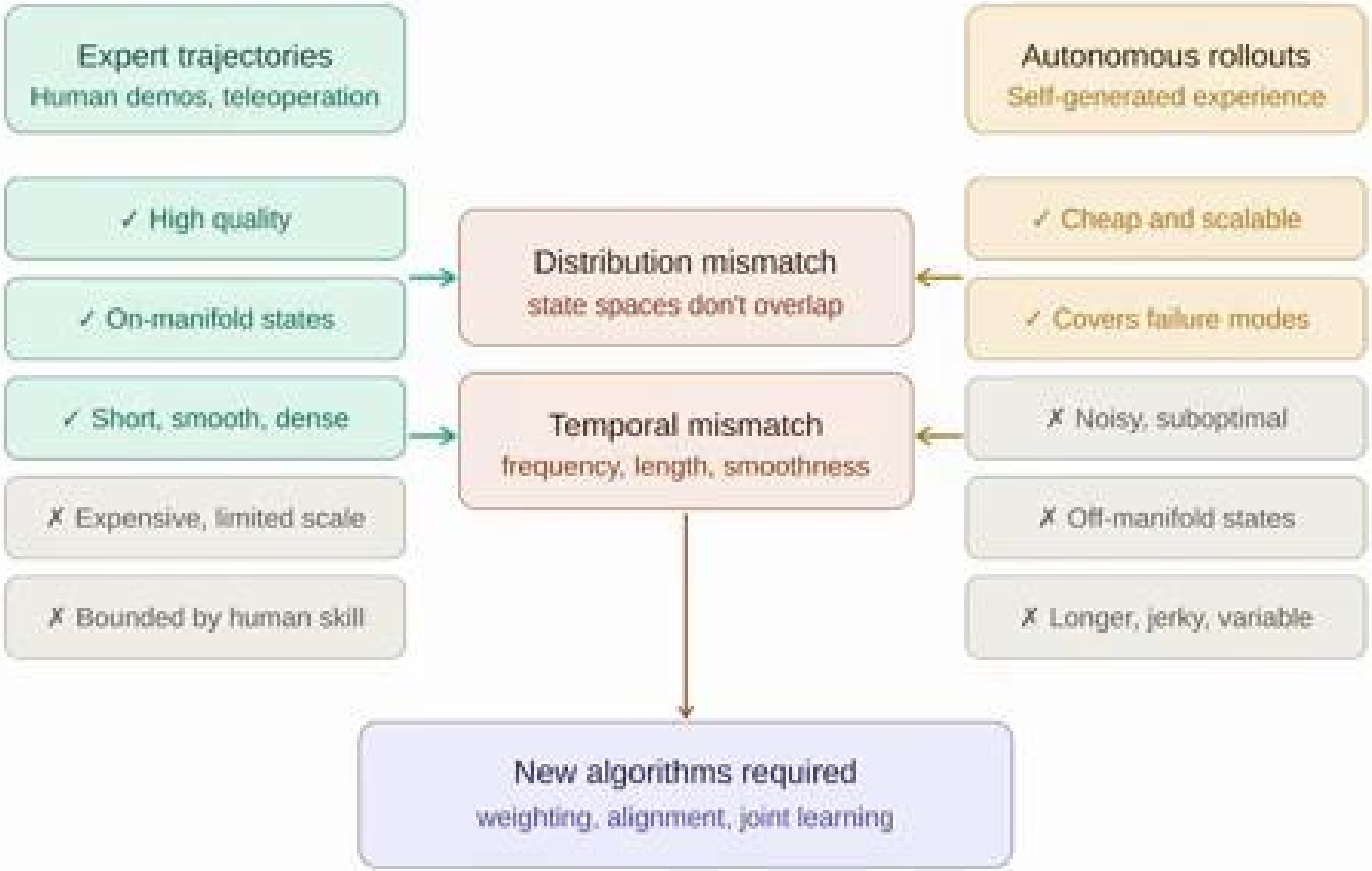


Offline-RL policy

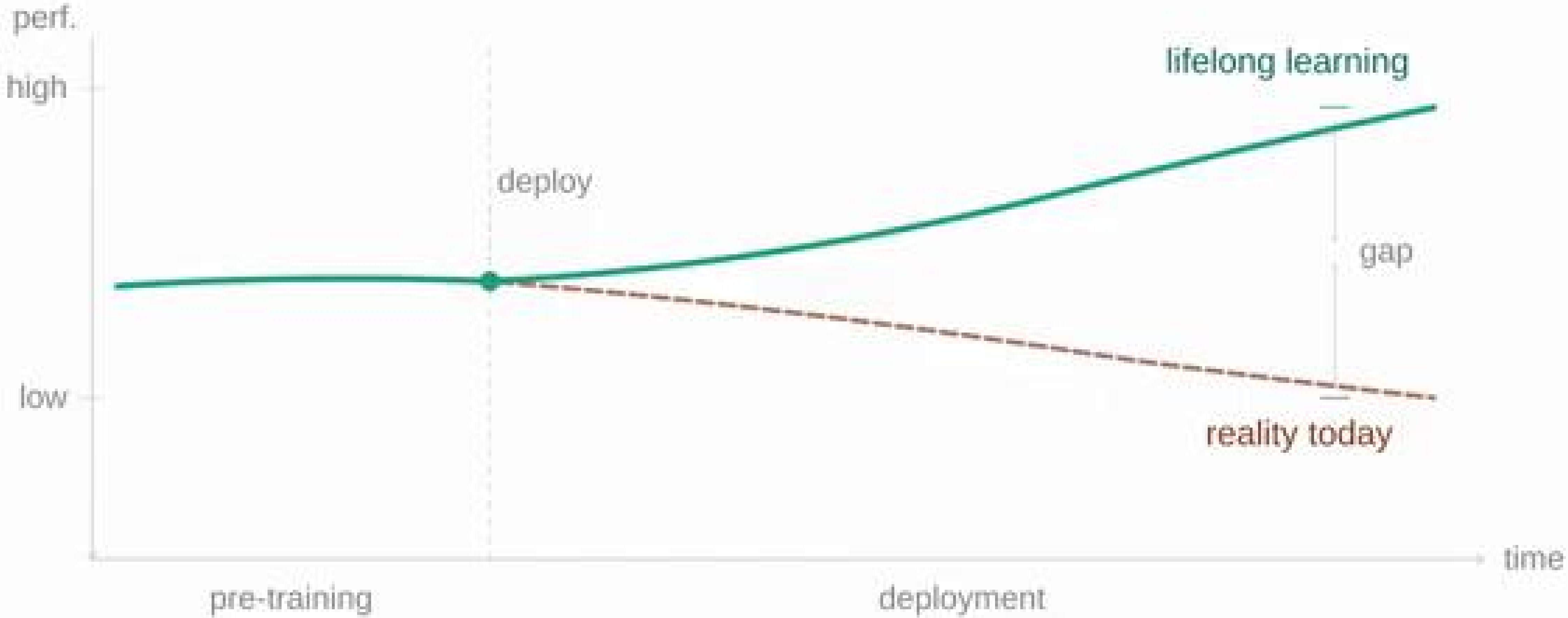


Robot Learning Data Flywheel

Co-Training Expert & Autonomous Data

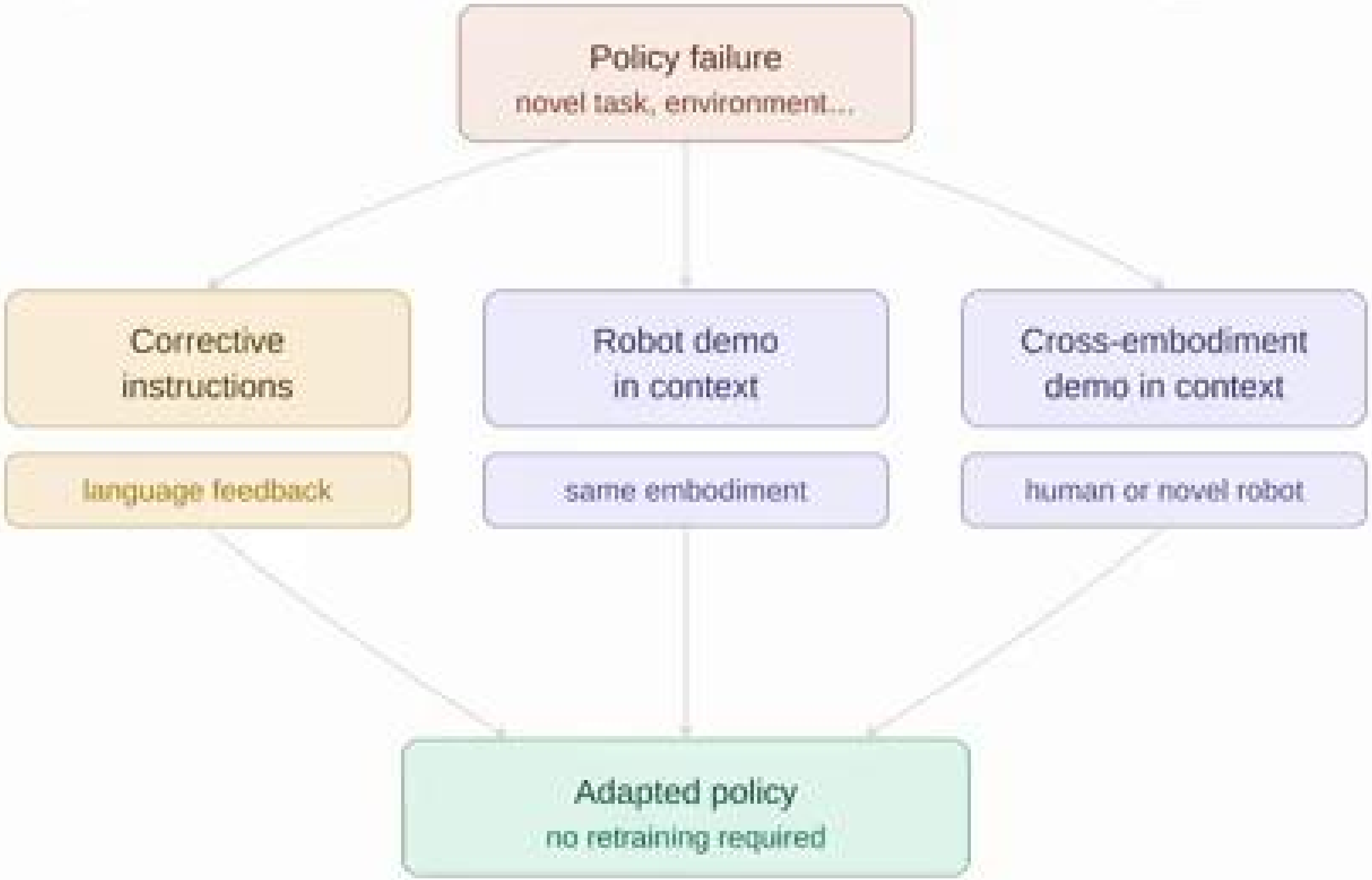


Lifelong Learning



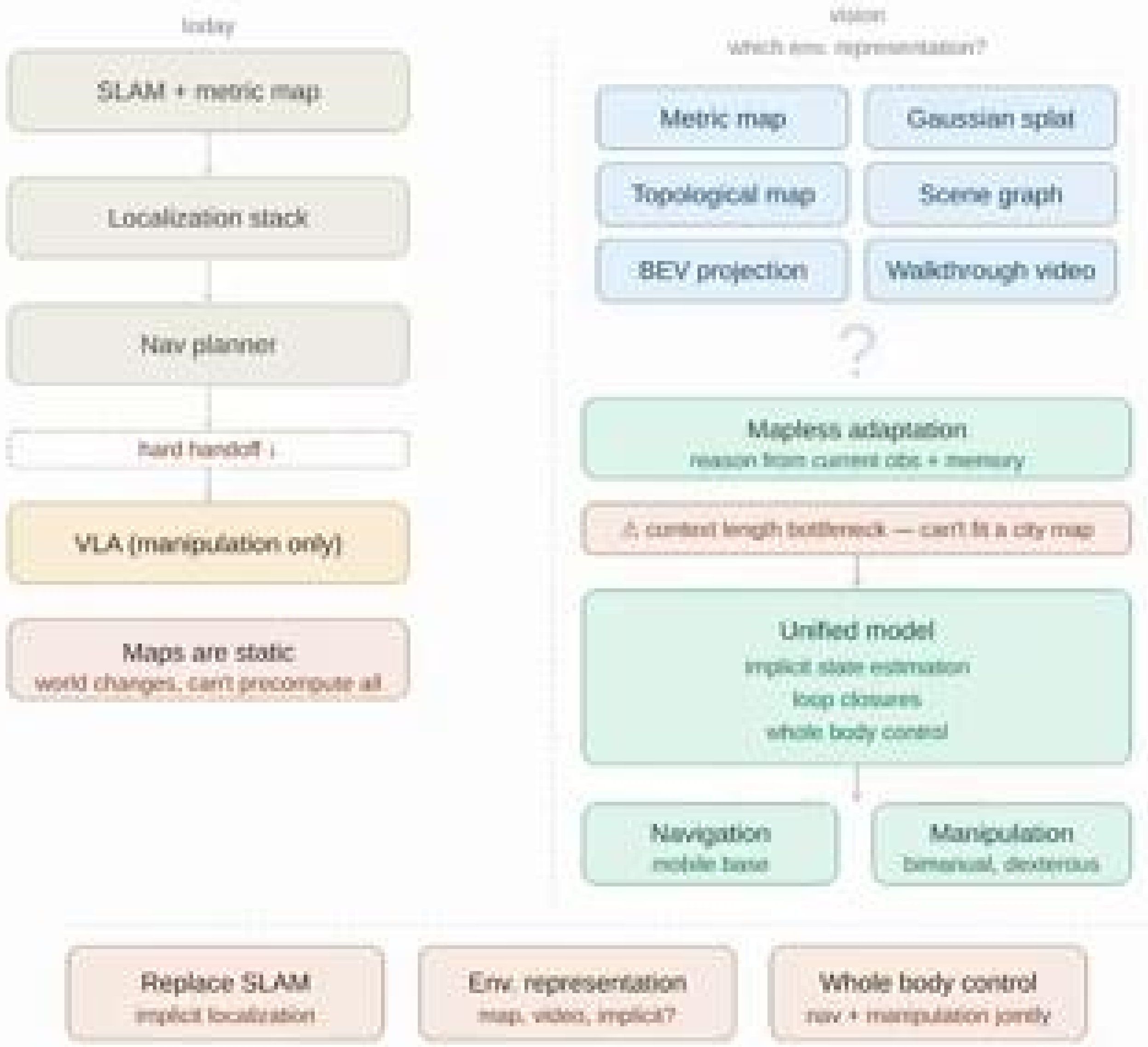
open problem: how do we get there?

Rapid Adaptation: In-Context Learning

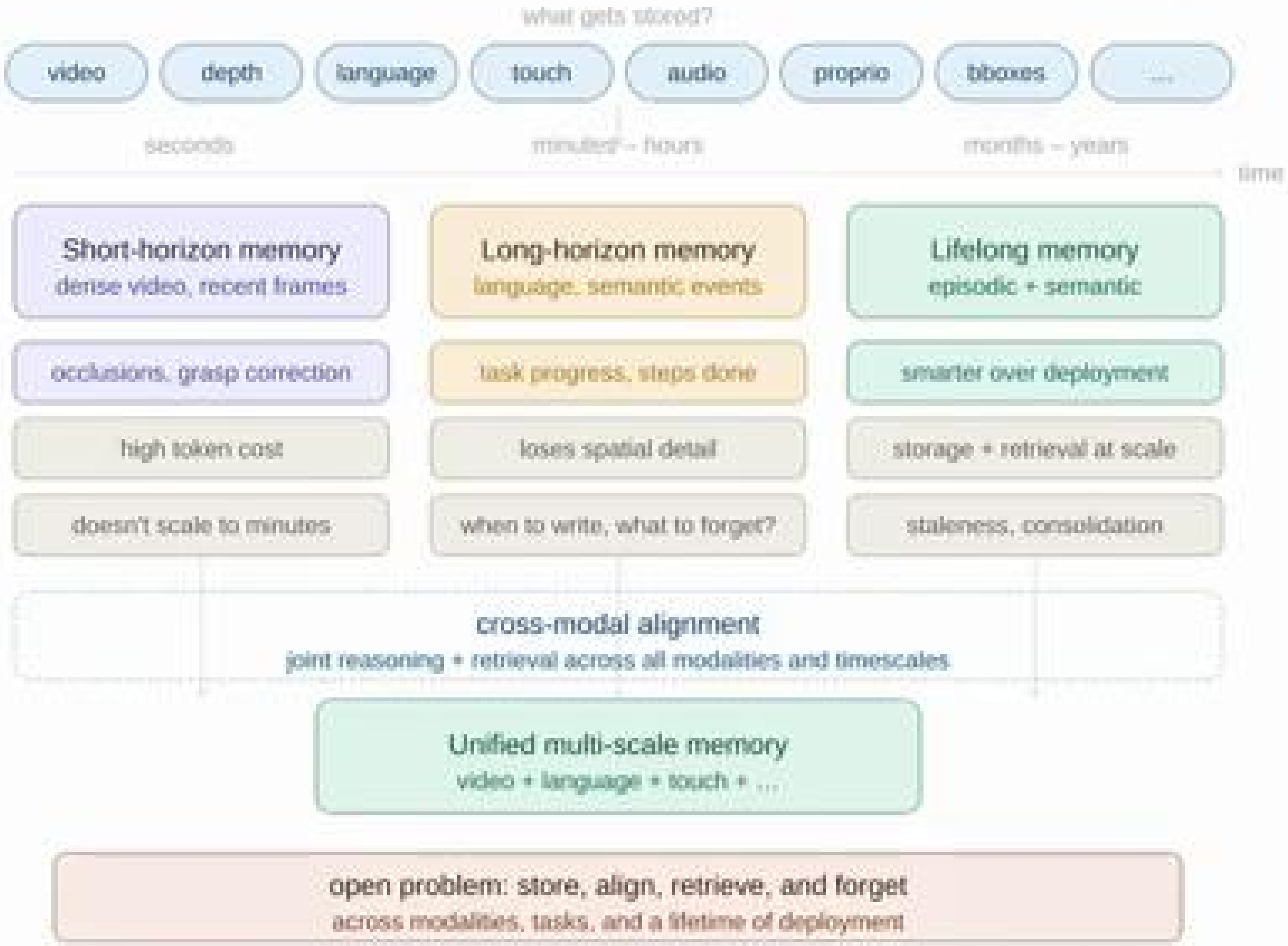


open problem: few-shot, real-time adaptation

Towards a Unified Model for Mobile Manipulation

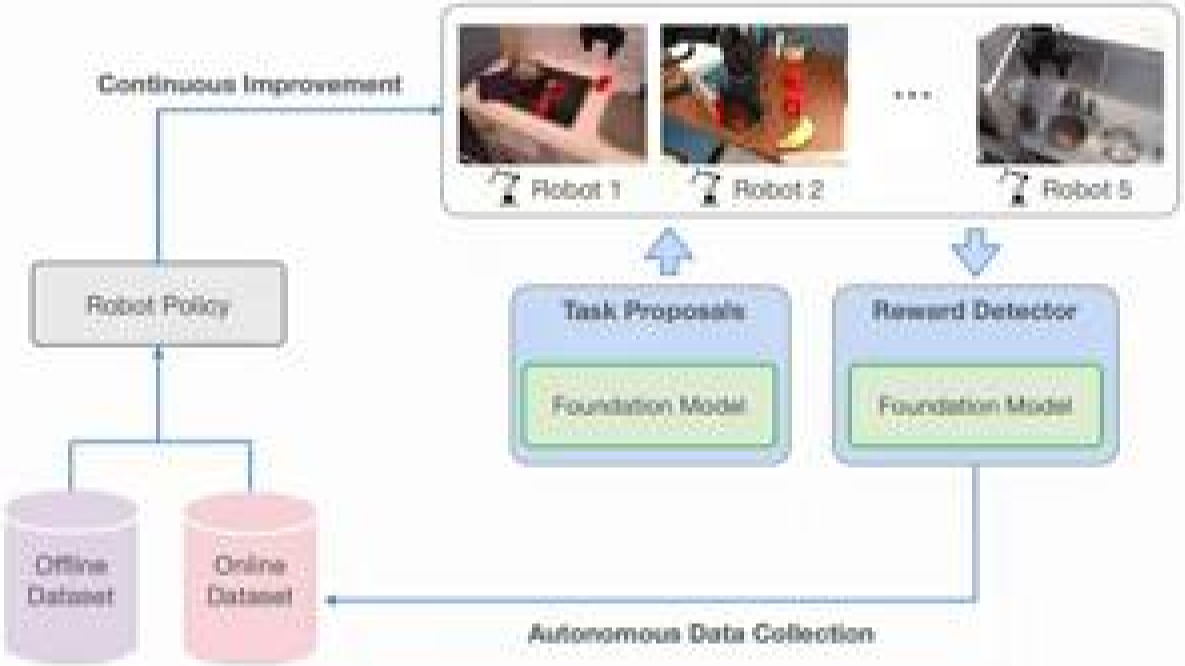


Multimodal Memories Across Time & Space



First Steps: Autonomous Improvement

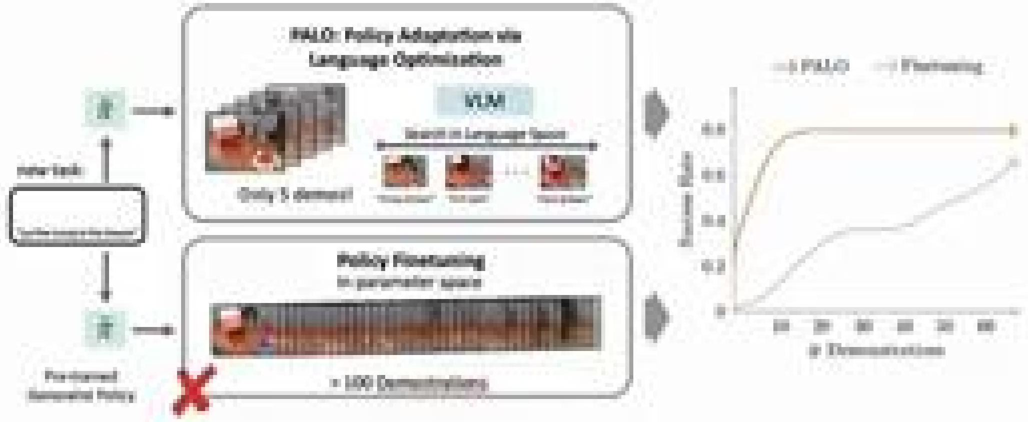
- Leverage foundation models to enable autonomous improvement without human interventions



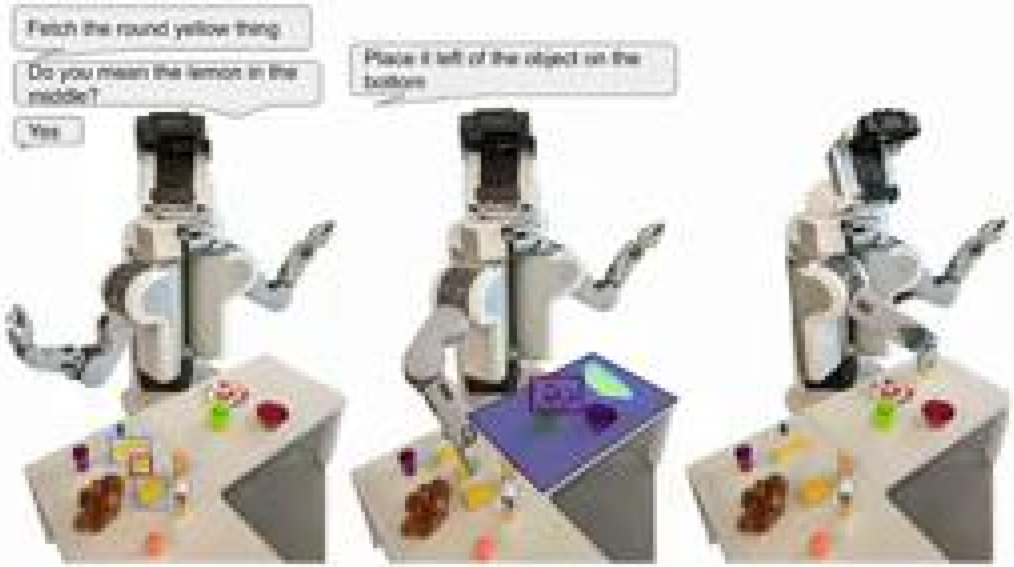
Autonomous Improvement of Instruction Following Skills via Foundation Models
Zhou*, Atreya*, Lee, Walke, **Mees**, Levine. CoRL, 2024.

First Steps: Adaptation via Human Feedback

- Leverage verbal or visual cues from humans to adapt to new tasks



Policy Adaptation via Language Optimization: Decomposing Tasks for Few-Shot Imitation
Myers*, Zheng*, **Mees** et al. CoRL 2024.



Composing Pick-and-Place Tasks By Grounding Language
Mees, Burgard ISER 2021.

Ingredients for Embodied Intelligence

Intelligent Embodied Reasoning

Dexterous Mobile Manipulation

Lifelong Learning

Rapid Adaptation

After this course, you have the tools to start advancing the frontier in robot learning!

But, how to do research in robot learning?

The (Harsh) Reality of Research

1. Most research is **incremental**

   today's self-driving cars rely on on 40 years of work:
   from Pomerleau's ALVINN in 1986 to the DARPA
   Challenge to modern end-to-end learning

2. Most research ideas never become papers.

3. Most papers don't stand the **test of time.**

4. The most impactful ideas are often the simplest,
   because **simple ideas can be scaled**

The Recipe for Good Research Problems

- Needed ingredients:
  1. an important problem
  2. a plan for how to tackle it

- Example ideas:
  - Cure cancer (important, but how?)
  - Algorithm that improves 1% on Libero benchmark (missing problem)

- If you are succesfull, **how does it help the community**?

The Recipe for Good Research Problems

- Needed ingredients:
  1. an important problem
  2. a plan for how to tackle it
  3. **excitement!**

Research requires tons of time and effort

You will be more likely to succeed if you are excited!

The Recipe for Good Research Problems

- Needed ingredients:
  1. an important problem
  2. a plan for how to tackle it
  3. excitement!
  4. be your own reviewer #2

What is the most likely reason your idea could fail?

If you can answer that honestly and still believe in the idea, proceed

Styles of Research

Method-driven

You start with an idea, but need to find the problem for it

"Video diffusion models just dropped, let me apply it to robot manipulation... somehow"

Problem-driven

You start with a problem, but need to find the method for it

"How can I make my VLA work with a novel camera viewpoint?

Both can lead to impactful research, but problem-driven is safer for a PhD student

Debugging Your Research

- Start with something that **should work**, then make it incrementally harder
- Talk to colleagues, authors of papers you are building upon, advisors etc.
- Visualize your model's data & outputs to understand its behavior
- Revisit your initial assumptions after experiments

Share Your Research

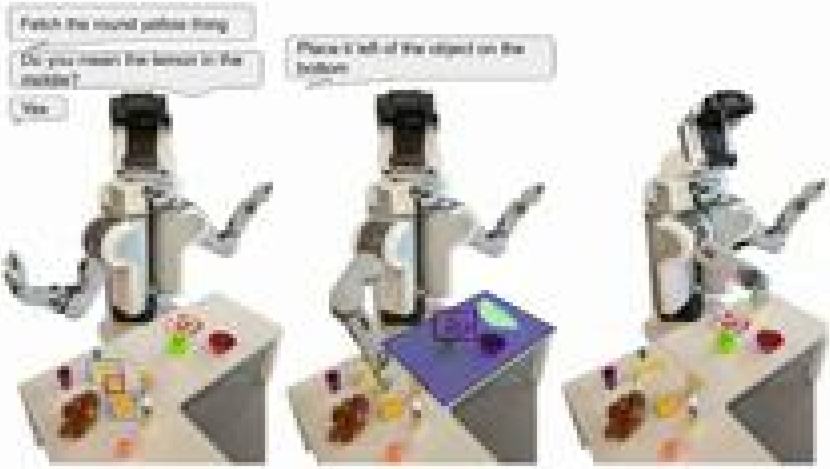
- Sharing your findings is how you **find your community**
- **Open-source** code & data: the most direct way for your research to be useful to others
- An image is worth a thousand words: polish your figures
- **Adapt** to your **audience**: even experts may know nothing about your specific topic

Share Your Research

- If nobody knows about your research, it didn't happen!
- Disseminate your research on social media
  - Find the line between making people curious about your work
  - And overhyping it like this:

Personal Backstory

Early PhD: frustrated with thousand ROS nodes, disjoint models and errors accumulating through a complex pipeline
Fun fact: robot broke a week before the deadline



- Can I ditch ROS and learn everything (motor skills+ language) end2end for a cool end of PhD demo?
- What would it take?
- Wrote a proposal that funded me for the next PhD years

Personal Backstory

Convinced my advisor to buy me a Franka robot, spent 1 year setting it up with a custom made table, VR control etc.

This was my last lecture of the course!

Thank you for your patience, this was a new course and your mid-term feedback was very helpful.

Teaching this has been an absolute privilege, even while juggling two full-time jobs.

I am looking forward to seeing your projects on demo day!

Huge Thanks to the Teaching Assistants

This course would not have been possible without them

References

- Uni Freiburg, Deep Learning Lab
- UC Berkeley Deep RL
- Stanford University, Deep RL
- Cornell Robot Learning